

Szegő's Theorem, Toeplitz Determinants, and the Pandrosion–Mahler Connection

Ivan Besevic

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Abstract

Szegő's theorem describes the asymptotic growth of Toeplitz determinants D_n from the geometric mean of the generating function f on S^1 . For $f = |P|^2$ where P is a polynomial, the Pandrosion–Jensen formula gives $G(|P|^2) = M(P)^2$ (Mahler measure squared), linking Toeplitz asymptotics to the Pandrosion field via Paper 59. The strong Szegő limit $D_n/G^n \rightarrow E$ is verified at 6-digit precision, with $E = 4/3$ for the single-root case. For $P = z^n - 1$, the Toeplitz matrix is diagonal (Paper 77 orthogonality), yielding $D_n = 2^n$.

1 Szegő's theorem

Theorem 1.1 (Szegő, 1920). *For $f > 0$ on S^1 with Fourier coefficients $c_k = \frac{1}{2\pi} \int f(e^{i\theta}) e^{-ik\theta} d\theta$, the Toeplitz determinant $D_n = \det[c_{j-k}]_{j,k=0}^{n-1}$ satisfies:*

$$\lim_{n \rightarrow \infty} D_n^{1/n} = G(f) = \exp\left(\frac{1}{2\pi} \int_0^{2\pi} \log f(e^{i\theta}) d\theta\right). \quad (1)$$

2 Pandrosion–Jensen: $G(|P|^2) = M(P)^2$

Proposition 2.1. *For a monic polynomial $P(z) = \prod (z - \alpha_j)$:*

$$G(|P|^2) = \exp\left(\frac{1}{\pi} \int_0^{2\pi} \log |P(e^{i\theta})| d\theta\right) = M(P)^2 = \prod_j \max(1, |\alpha_j|)^2. \quad (2)$$

This follows from Jensen's formula (Paper 59): the log-integral of $|P|$ on S^1 equals $\log M(P)$. Verified: $G(|P|^2) = M(P)^2$ for 100 random polynomials, 0 violations.

3 The strong Szegő limit

Theorem 3.1 (Strong Szegő). *Under regularity conditions:*

$$\lim_{n \rightarrow \infty} \frac{D_n}{G(f)^n} = E(f) = \exp\left(\sum_{k=1}^{\infty} k |\hat{c}_k|^2\right), \quad (3)$$

where $\hat{c}_k$ are the Fourier coefficients of $\log f$.

n	D_n/G^n	E (theory)	ratio
$P = z - 0.5, G = 1, E = 4/3$			
3	1.328125	1.333333	0.996
5	1.333008	1.333333	1.000
8	1.333328	1.333333	1.000
12	1.333333	1.333333	1.000
$P = (z - 0.3)(z + 0.3), G = 1$			
3	1.016332	1.016399	1.000
8	1.016399	1.016399	1.000
16	1.016399	1.016399	1.000

The Szegő constant $E = 4/3$ for $P = z - 1/2$ is exact. Convergence is achieved by $n = 8$.

4 Cyclotomic case: diagonal Toeplitz

For $P = z^n - 1$ (Paper 77: Gram orthogonality), $f = |z^n - 1|^2$ has Fourier coefficients:

$$c_k = \begin{cases} 2 & k \equiv 0 \pmod{n} \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

The $n \times n$ Toeplitz matrix is $T = 2I_n$, giving $D_n = 2^n$. This connects Szegő asymptotics directly to the Pandrosion Gram orthogonality of Paper 77.

5 Pandrosion dictionary

Szegő theory	Pandrosion tool
Generating function f	$ P ^2$ on S^1
Geometric mean $G(f)$	$M(P)^2$ (Mahler, Paper 59)
Log-integral	Jensen via F_P
Strong Szegő constant E	Fourier of $\log P ^2$
Diagonal Toeplitz	Gram orthogonality (Paper 77)
$D_n^{1/n} \rightarrow G$	Root distribution on S^1

References

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